Anomaly Detection in 5G Networks using Federated Learning

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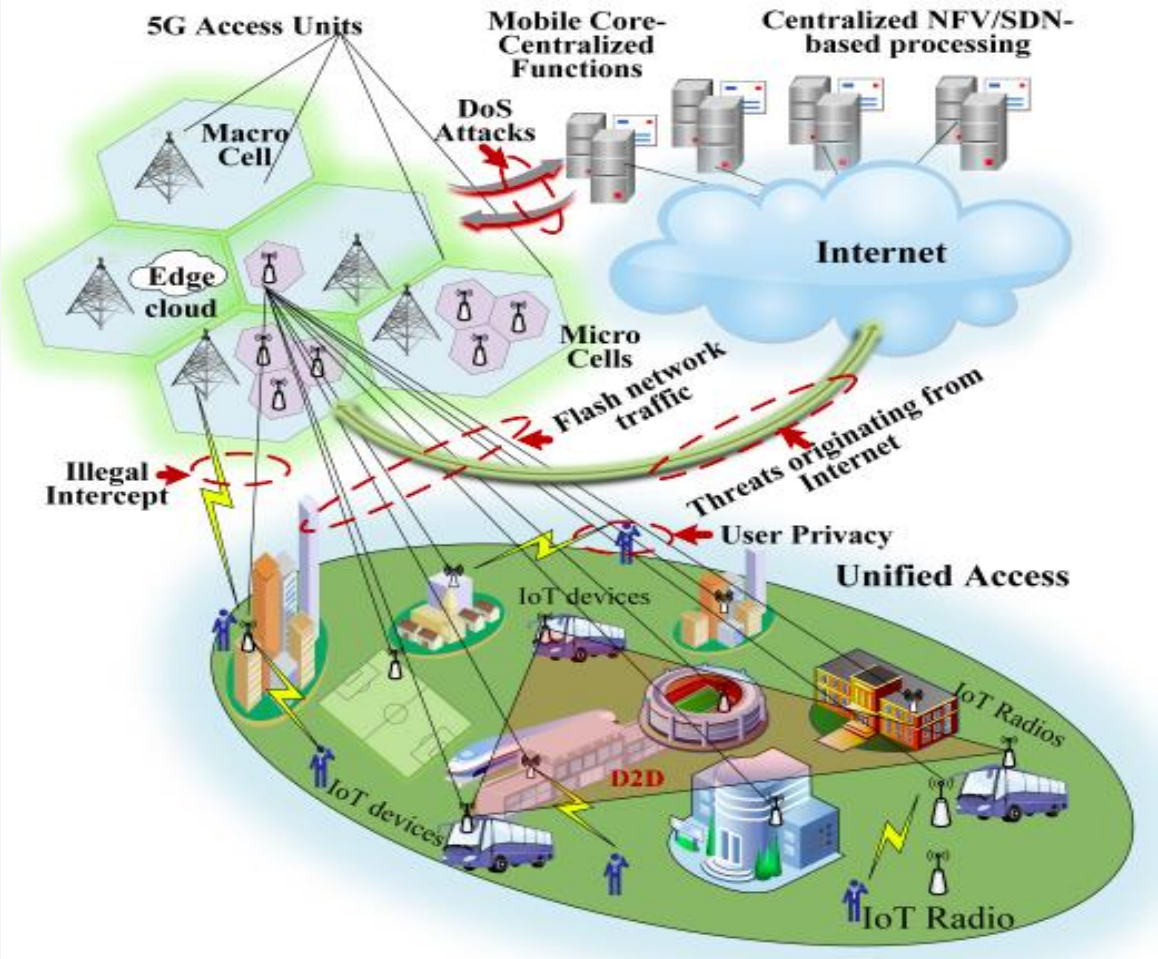
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OUTLINE

1. Introduction
2. Problem Statement
3. Literature Review
4. Proposed Solution & Objectives
5. Methodology
6. Conclusion

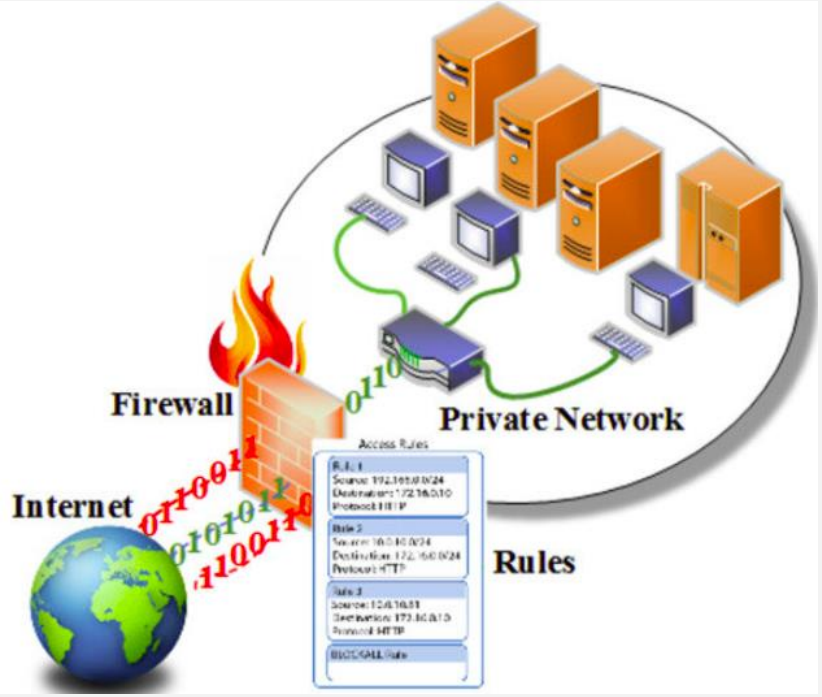
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INTRODUCTION

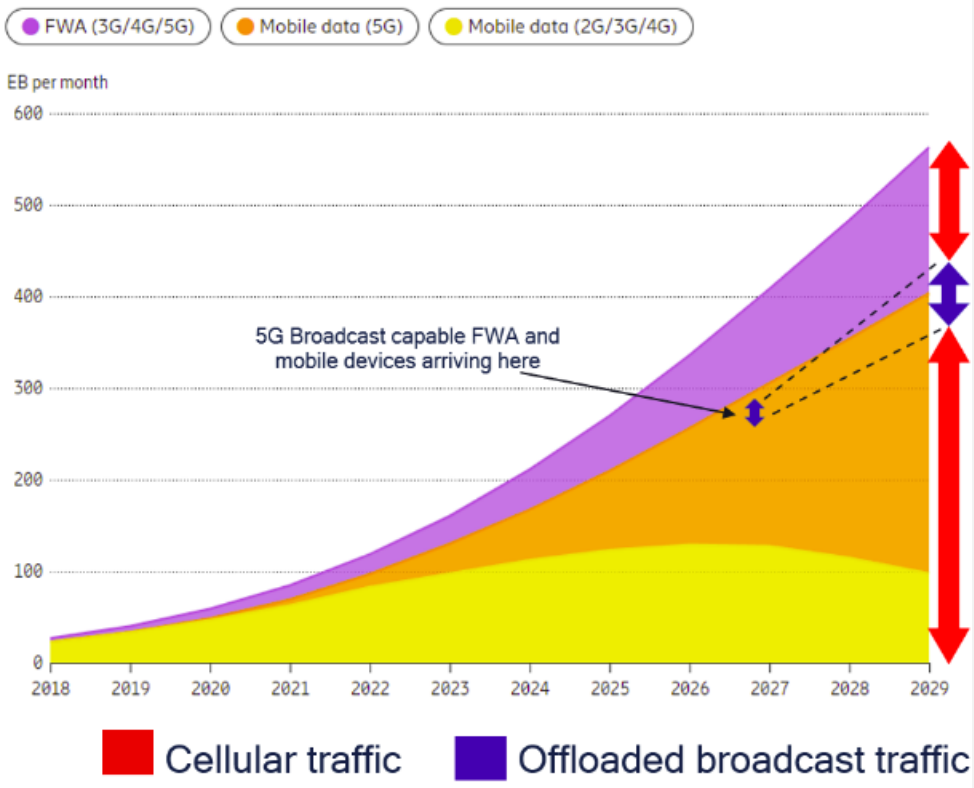


5G network and the threat landscape.

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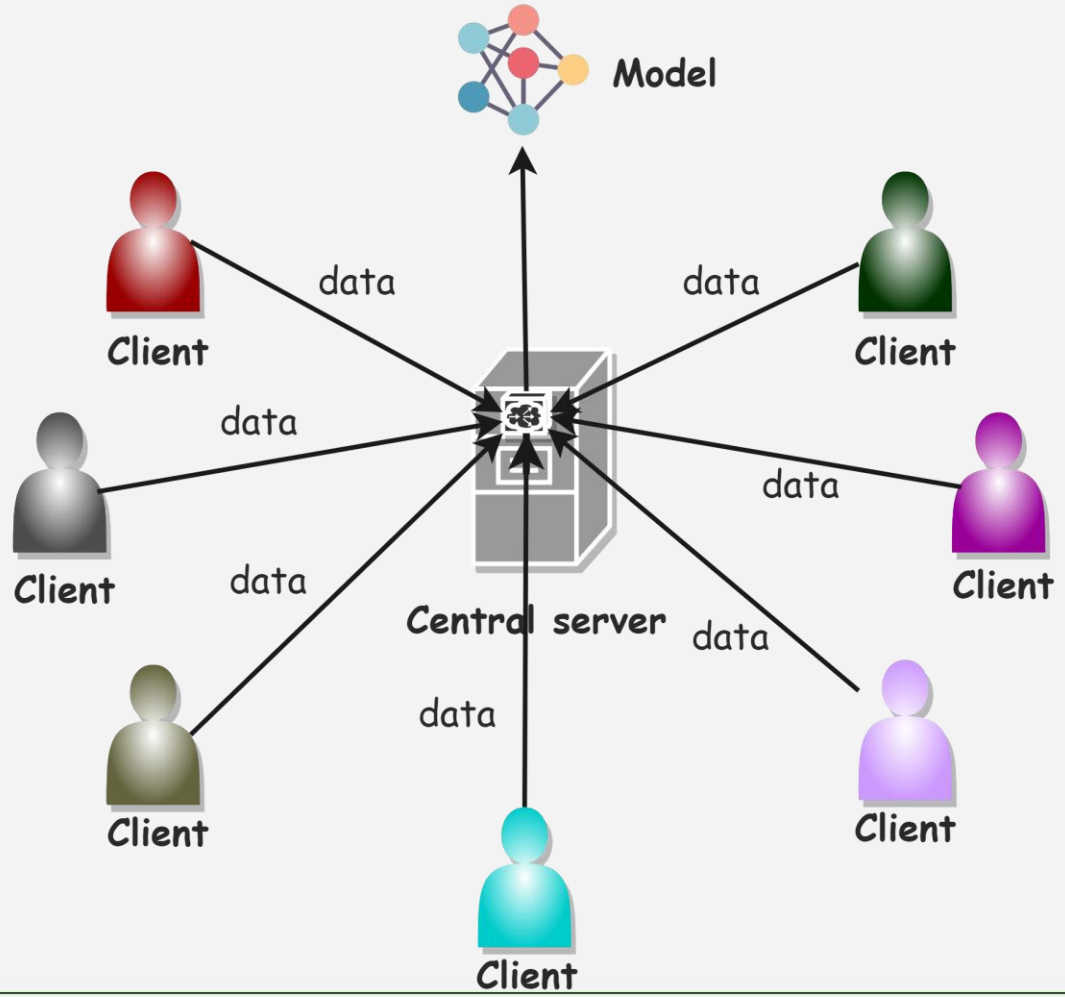
TRADITIONAL IDS



DATA EXPLOSION

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TRADITIONAL MACHINE LEARNING

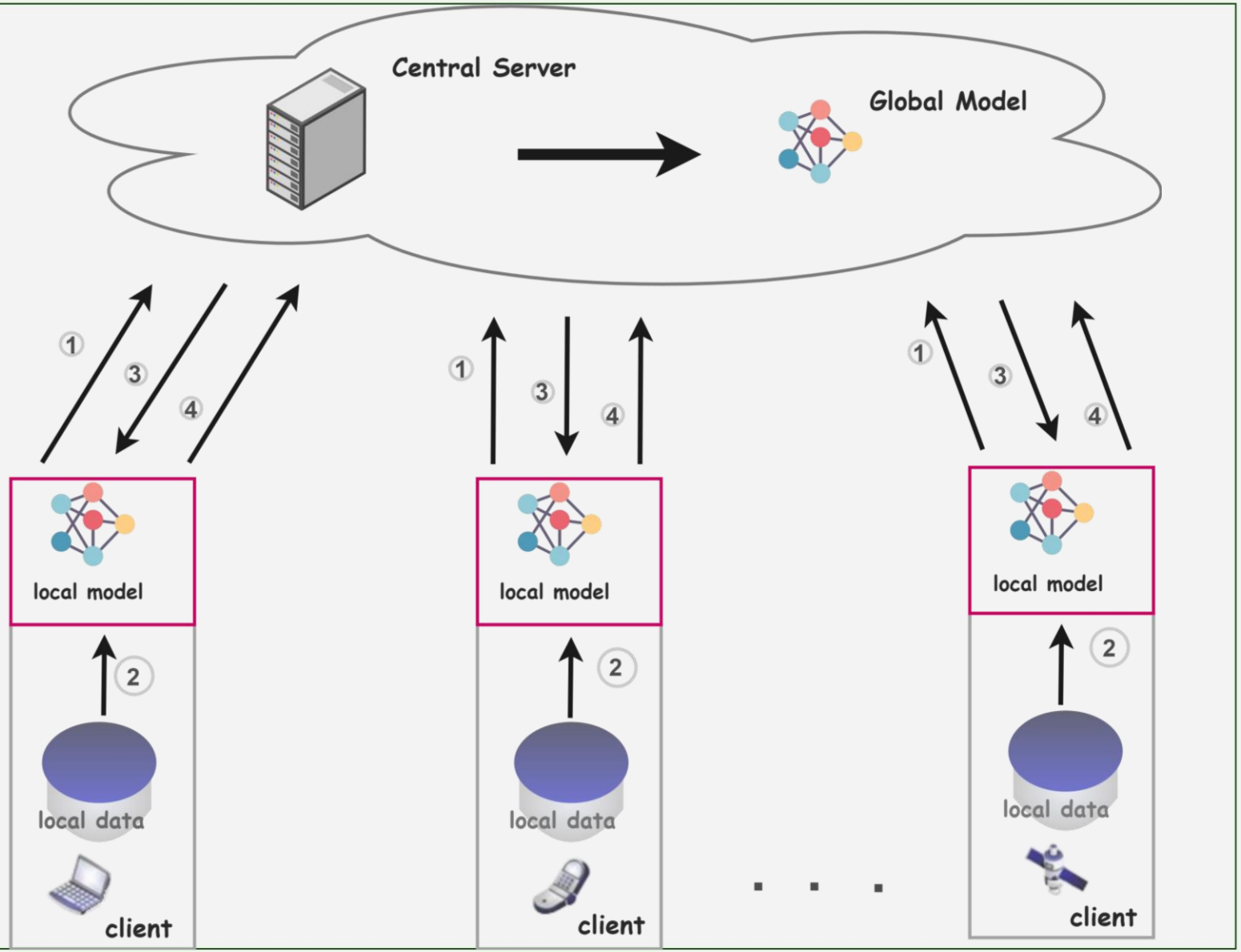


PROBLEM STATEMENT

Traditional anomaly detection models in 5G networks often involve a centralized approach, where client data is transmitted to a central server, raising significant security and privacy concerns.

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FEDERATED LEARNING



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LITERATURE REVIEW

RESEARCH PAPERS	METHODOLOGY	FINDINGS	DRAWBACKS
4G and 5G: Security and privacy analysis(2020), Omerani Hanane, Mazri Tomader	Used the combination of - literature review - comparative analysis of security features and vulnerabilities in 4G and 5G networks -examined existing security mechanisms to evaluate the effectiveness of the security measures	- Established the threats landscape of 4G and 5G.	Failed to propose solutions to the security flaws in 5G.
Network Anomaly Detection in 5G Networks(2022), Atta Rahman Tahr Iqbal, Maqsood Mahmud, Linah Saraireh.	- Merged clustering methods, K-prototype, KNN classification method, and novel clustering-filtering-refinement (Integrated Method). - Compared it to a standalone K-prototype and standalone KNN classification method for unprecedented anomaly detection in network logs.	- The integrated method outperformed the K-prototype and the KNN classification method in terms of false alarm rate and precision rate	Data privacy and computational resources concerns.
Deep Learning in 5G Wireless Networks - Anomaly Detections(2020), Minh Doan, Zhanyang Zhang.	- Transformed the network flow data to images and from the original U-Net architecture to Res-UNet architecture and used CTU datasets to evaluate the result of the model.	-Showed that Res-UNet can be an alternative deep learning approach for self-adaptive anomaly detection in 5G networks. - Compared to the commonly used DBN and LSTM, Res-UNet provided an advantage of classifying time-dependent features in anomaly detections without heavy delay	Data Privacy The datasets were highly imbalanced.

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LITERATURE REVIEW CONT'D

RESEARCH PAPERS	METHODOLOGY	FINDINGS	DRAWBACKS
A 5G Slice Traffic Anomaly Detection Method Based on Convolution Neural Network(2023), Chen Lu, Zhang Tao, Ma Yuanyuan	- Converted 5G slice traffic into images for analysis by a CNN model. - Used NAS (Neural Architecture Search) to detect anomalous network traffic using the IDS2018 dataset. - Improve the accuracy and efficiency of the intrusion detection system with SDS (Software Defined Security)	- Compared the performance of the CNN model with a traditional model. - The accuracies of both traditional binary and multiclass classifications were significantly lower than that of CNN-based detection accuracy.	Data privacy
Network Anomaly Detection Using Federated Learning(2023), William Marfo, Deepak K. Tosh, Shirley V. Moore	- Used ANN to detect anomalies in 5G using HFL approach and UNSW-NB15 dataset	- Results showed that the UNSW-NB15 dataset improved the performance of ANNs on client devices to detect anomalous requests when trained in the federated learning environment - In comparison to when they were trained using classical machine learning models.	Faced data imbalance that affected the accuracy performance.
Secure Federated Learning in 5G Mobile Networks(2020), Martin Isaksson, Karl Norrman	- Designed a multi-party computation scheme for aggregation in a privacy-preserving fashion. - Integrated the scheme into the NWDA framework without breaking the structure.	Their findings showed that integrating FL into NWDA architecture reduced sensitive information leakages and significantly improves privacy	Their scheme had a high communicational overhead

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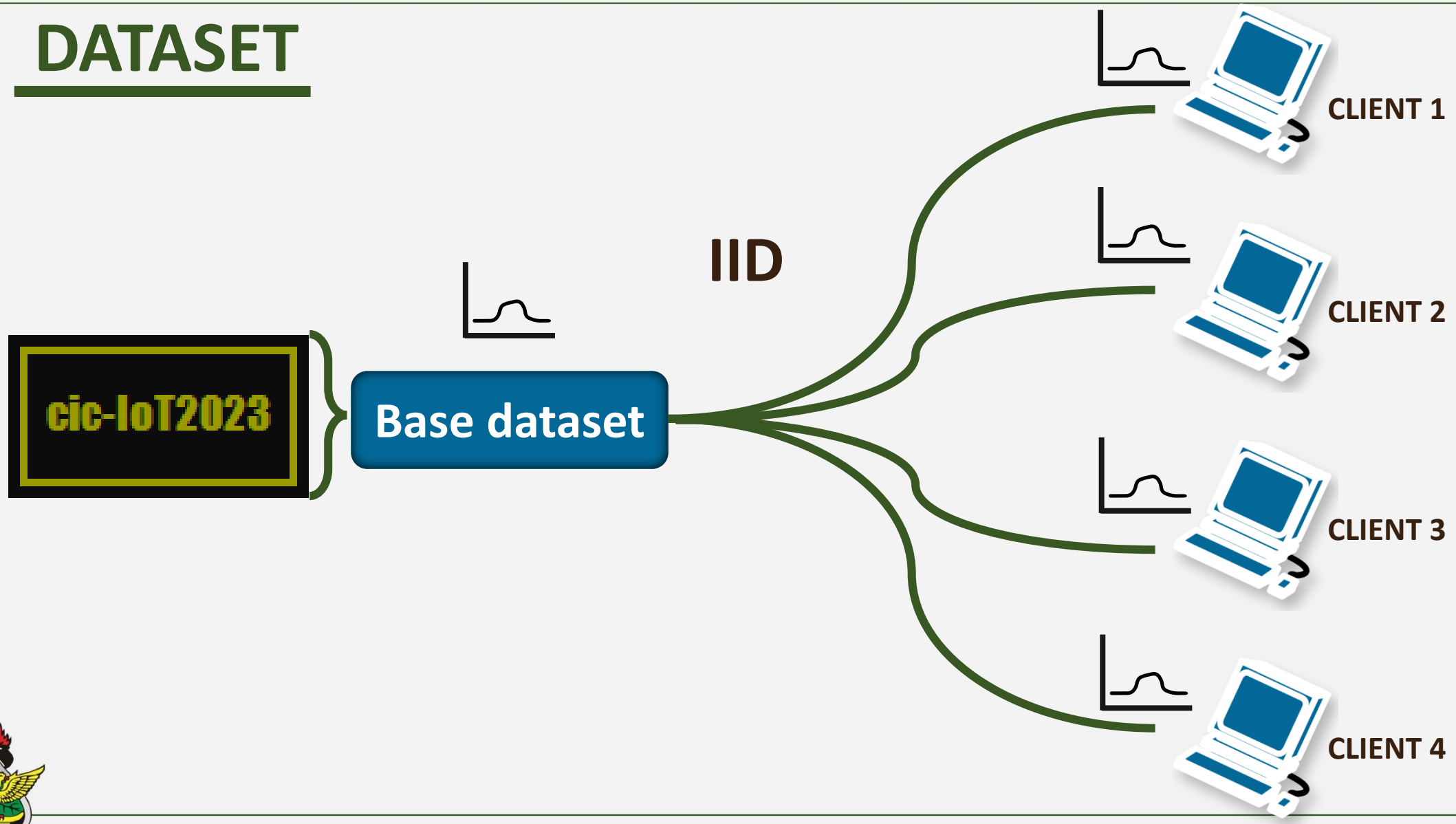
PROPOSED SOLUTION AND OBJECTIVES

To create a federated learning model for anomaly detection in 5G networks using IID.

- Evaluate the performance of the model developed by a classical deep neural network
- Perform hyperparameter tuning to improve the performance of the model.
- Evaluate the performance of the model developed by federated learning.
- Perform hyperparameter tuning to improve the performance of the model.
- Comparative analysis of Classical Deep Learning and Federated learning technique.

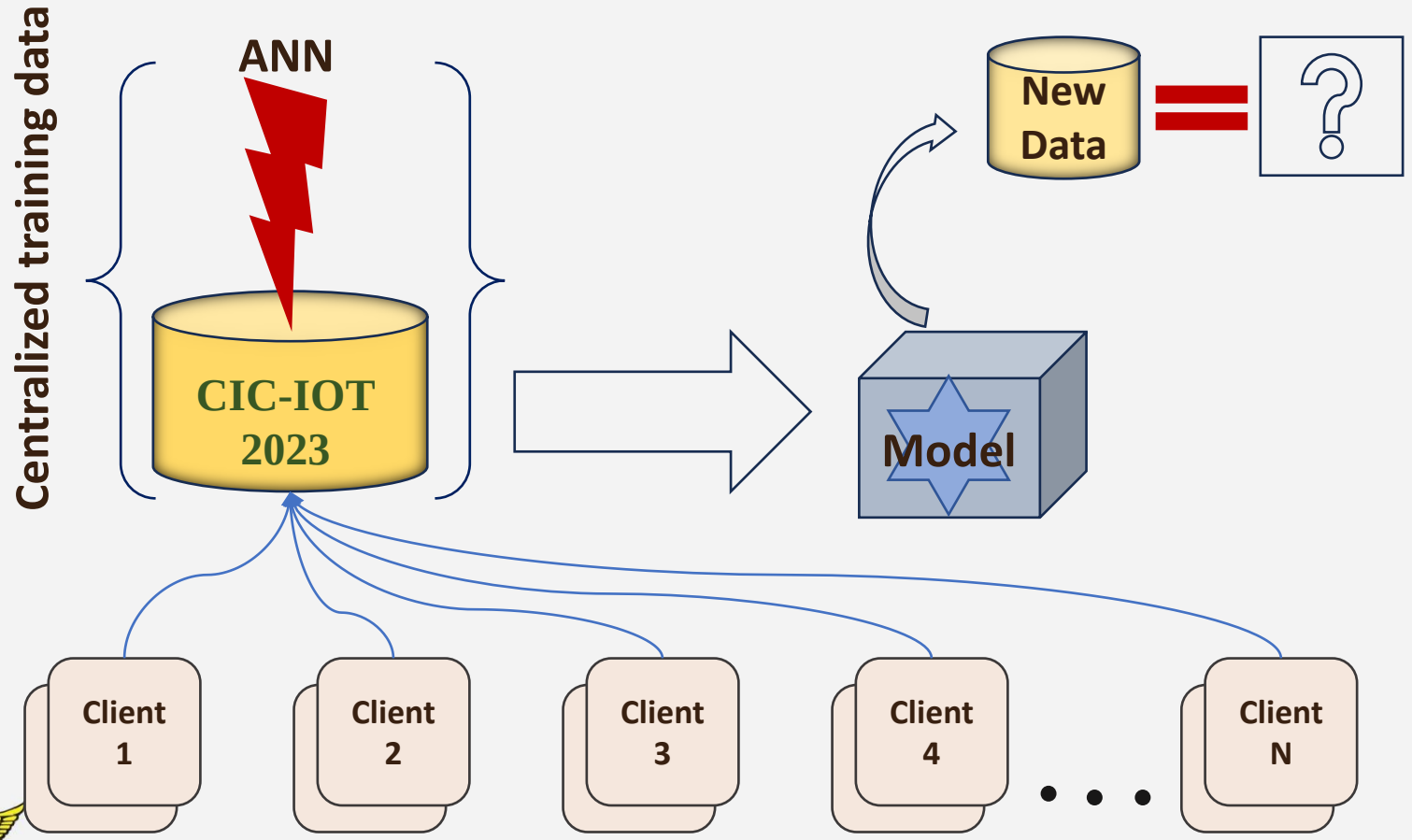
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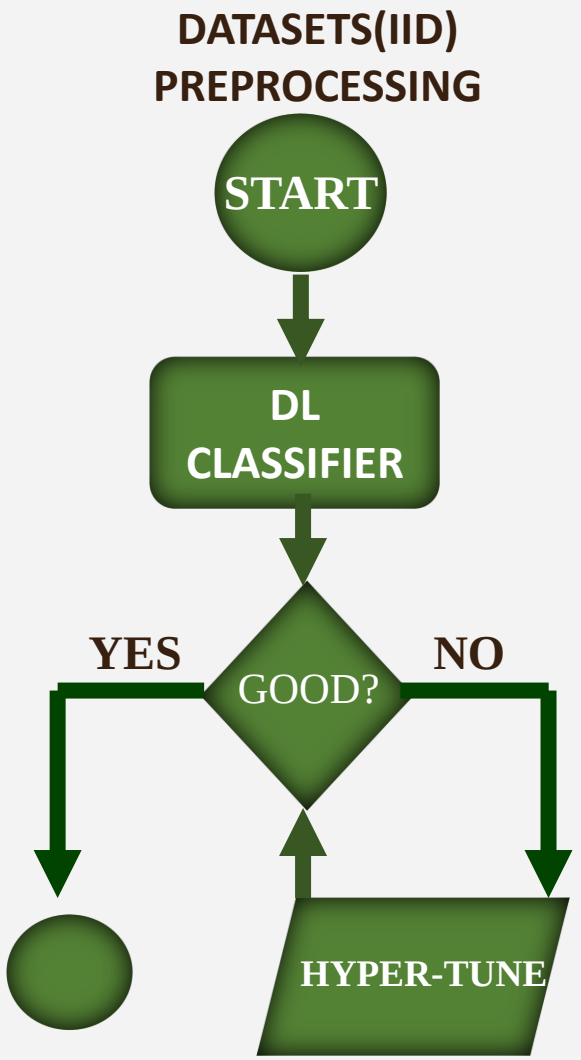
DATASET



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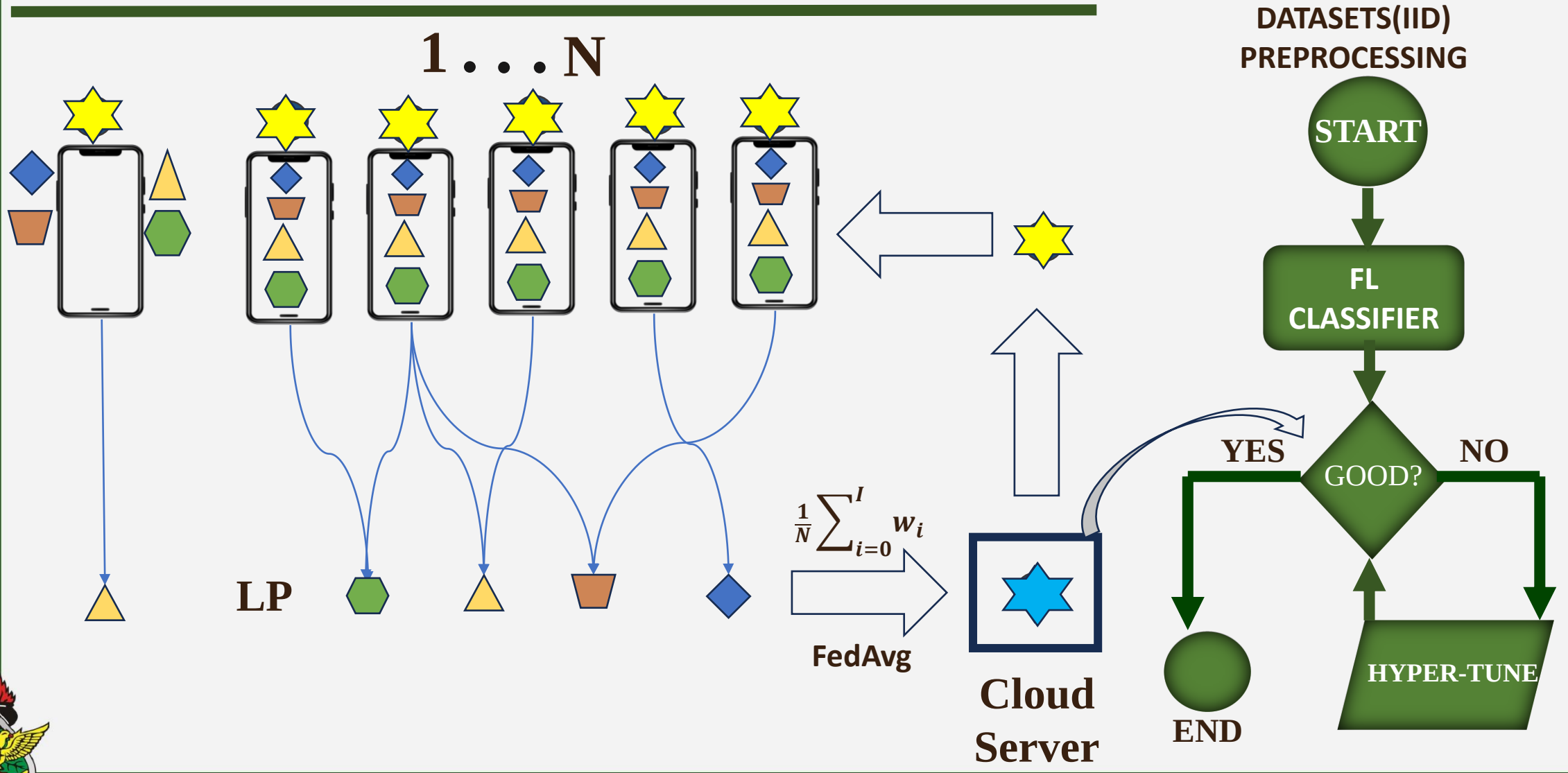
METHODOLOGY: TRADITIONAL ML





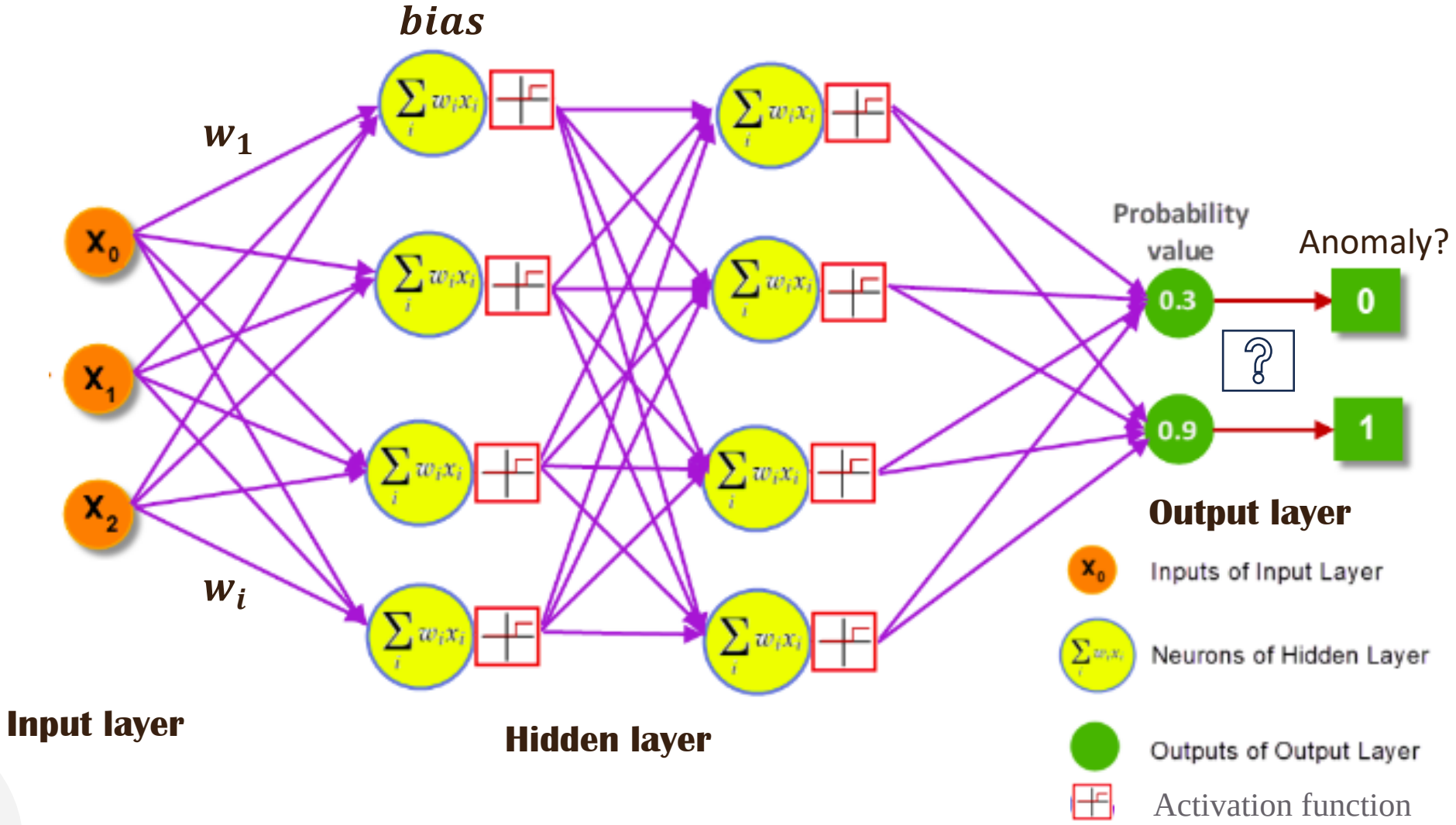
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METHODOLOGY: FEDERATED LEARNING



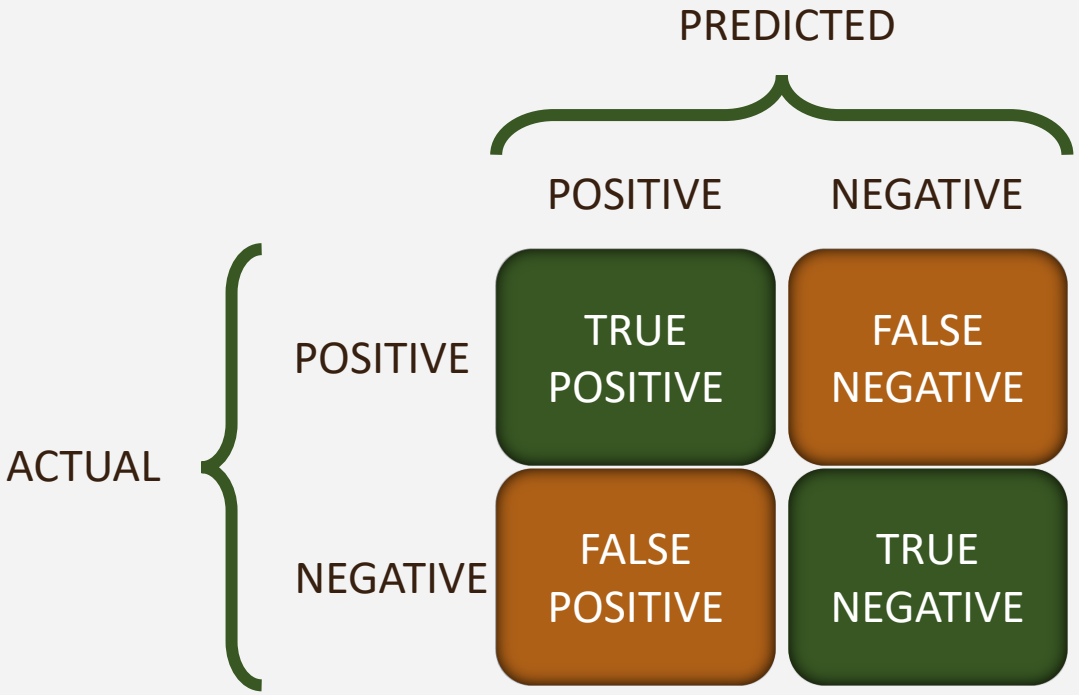
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ARTIFICIAL NEURAL NETWORK: ARCHITECTURE



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PERFORMANCE METRICS



CONFUSION MATRIX

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+TN}$$

$$F1\ score = 2\,x\,\frac{Precision\ x\ Recall}{Precision+Recall}$$

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

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TOOLS NEEDED

Python



Tensor Flow



Flower

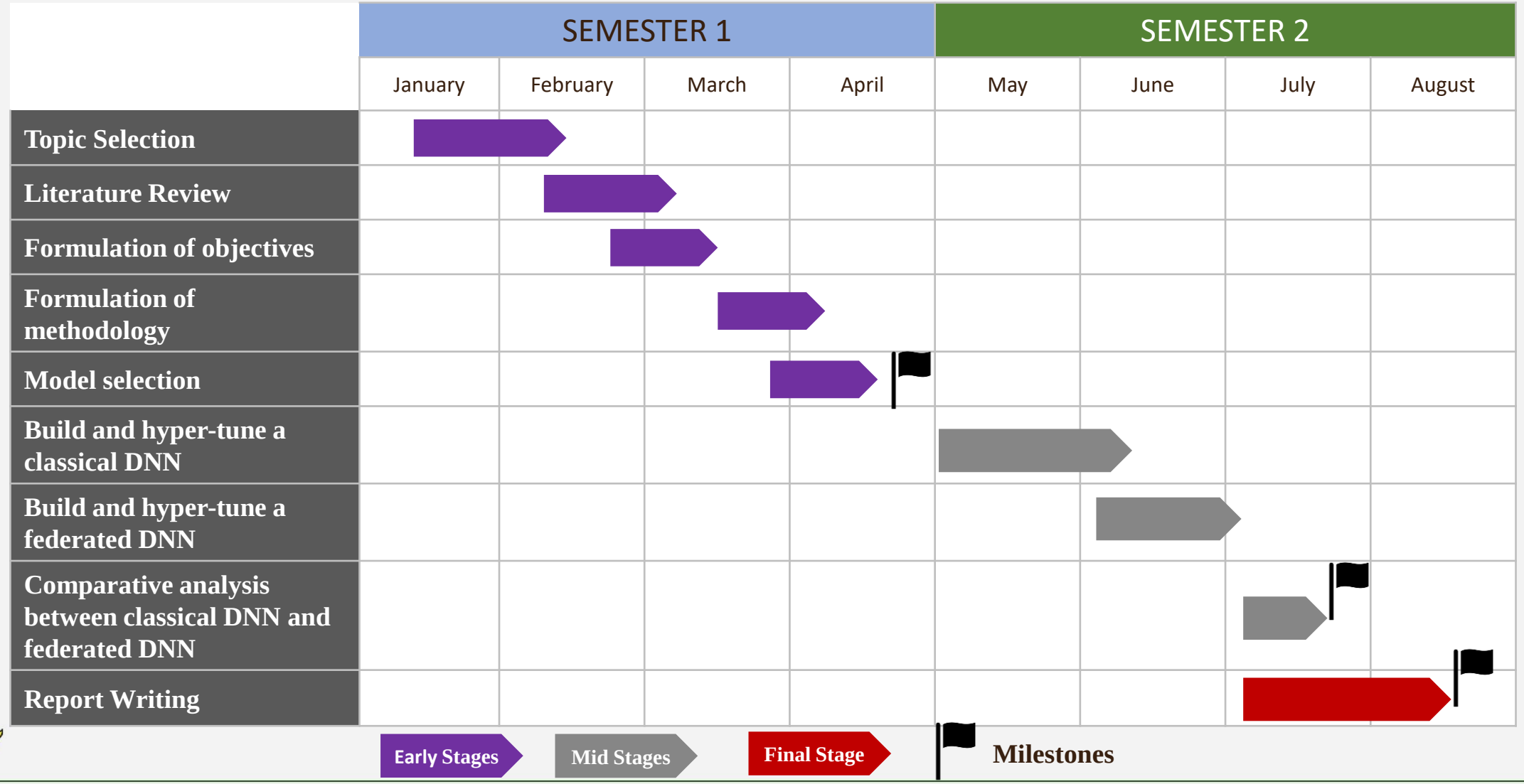


TF Federated



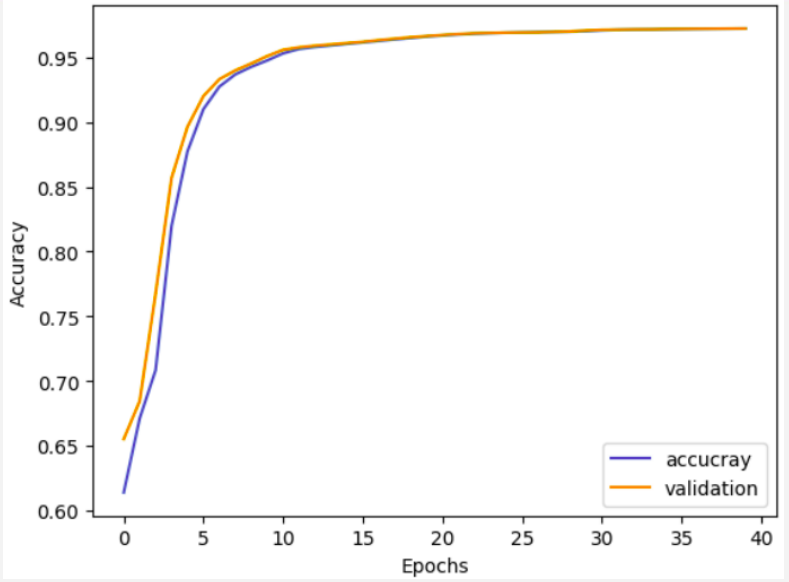
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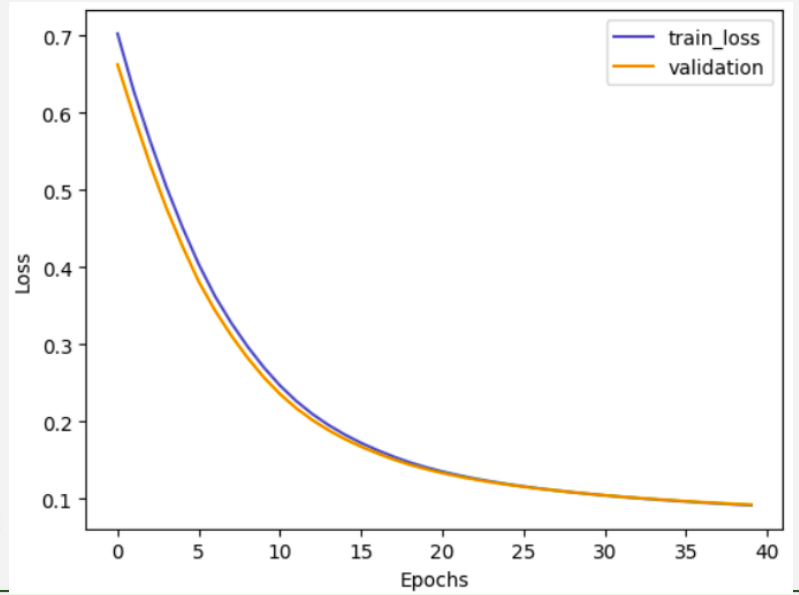
PROJECT TIME LINE



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Progress with the classical deep neural network





Model's performance on unseen data

Confusion Matrixs

```
[[103172    337]
 [  4472  68584]]
```

Evaluation metrics

	precision	recall	f1-score	support
0	0.96	1.00	0.98	103509
1	1.00	0.94	0.97	73056
accuracy			0.97	176565
macro avg	0.98	0.97	0.97	176565
weighted avg	0.97	0.97	0.97	176565

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CONCLUSION



5G Networks



FL System



Contribution

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